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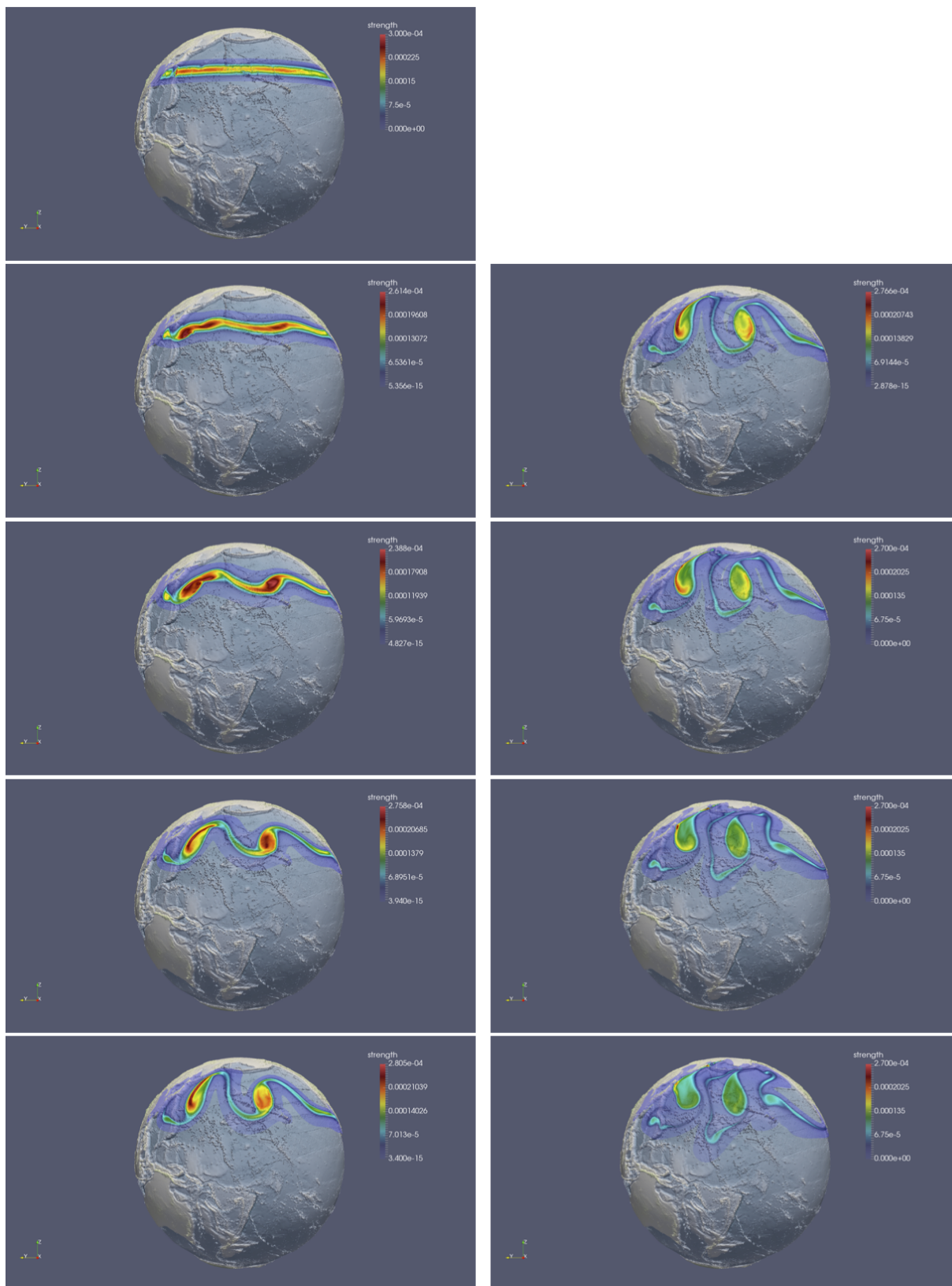
A New Vortex-in-cell Method for Solving the Shallow Water Equations on the Sphere

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Abstract Text:

We present a new vortex-in-cell method for computational modeling of the quasi-geostrophic shallow water equations on the sphere. We assume that, ψ , the stream function and, h_s , the deviation of the water free surface from its resting position are related by the local linear balance, $g h_s = f \psi$, following the arguments by Kuo (1959) and Charney and Stern (1962), where g is gravity and f is the Coriolis parameter. For small h_s compared to the water depth $d(\varphi, \theta)$, the Global quasi-geostrophic shallow water equation is expressed as $\nabla^2 \psi - \psi/L_d^2 \approx Q d - f$ (eq 1), where the Rossby radius is $L_d = (gd(\varphi, \theta))^{1/2}/f(\varphi, \theta)$, and $Q(\varphi, \theta, t)$ is the potential vorticity, which is conserved along particles trajectories. Spatio-temporal variations in the water depth $h(\varphi, \theta, t)$, contribute to an irrotational component of the velocity, with the potential, Φ , governed by Poisson's equation $\nabla^2 \Phi = -h^{-1} Dh/Dt$, where D is the Lagrangian derivative and $h(\varphi, \theta, t) = h_s(\varphi, \theta, t) + d(\varphi, \theta)$. In the proposed hybrid method, Lagrangian particles carrying the potential vorticity are advected using a second order Runge Kutta time integration scheme. The velocity at the particles locations is interpolated from the velocity field computed on the grid using Grid-based Poisson's solvers for ψ and Φ . Assessment of the proposed method is carried out by exploring its accuracy as a function of the

grid size and time step and by comparing with existing Vortex-in-cell methods.



Plain-Language Summary:

We are proposing a new computational model for the dynamic simulation of the shallow water waves on the sphere. The proposed method combines the strengths of efficient grid-based solvers of the Poisson and Screened Poisson solvers and of Lagrangian method that efficiently handle the

fact that the potential vorticity is conserved along particles trajectories. The method account for variation in the bathymetry in longitude and latitude.

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